DESIGN PLAN · 28 OCTOBER 2026

A pre-map trading model for Valorant

Player-based round Elo, a map layer and a veto model, priced against Kalshi's Valorant series and map-winner markets. Built on the Prometheus playbook; the website comes later.

Short answer: yes. Kalshi already lists Valorant series markets (KXVALORANTGAME, 1,673 matches since Sep 2025) and per-map winner markets (KXVALORANTMAP, 3,316 map markets since Feb 2026), across tier 1, Challengers/VCL and Game Changers. VLR.gg publishes everything an Elo model needs, for free: vetoes, map scores, halves, round outcomes, economy and agents. Much of Prometheus carries over: the Elo engine, benchmark rules, Kalshi plumbing, sizing and executor. What's new is the data scraper, round-level ratings, maps, and the veto.

Your answers that shaped this plan

Question	Answer	What it means here
Pre-map scope	Both	Series price before the veto; map 1 and 2 prices once the veto is public; re-price between maps.
Venue	Kalshi	Reuse Prometheus's Kalshi code; trade KXVALORANTGAME + KXVALORANTMAP.
Data budget	Free / scraped	VLR.gg primary, bo3.gg cross-check, with an adapter layer so a paid feed can be swapped in.
Leagues	All, incl. Game Changers	Tier 1, Challengers/VCL, Ascension, GC and third-party events. Kalshi lists all of them.
Backfill	2023+ scored, 2021–22 seeds	Ratings start in 2021; benchmarks score 2023 on.
Third-party events	Include and trade	EWC, POKAL, Project Blender etc. feed the ratings and are traded like any other event.
Bankroll	Same limits as League; $50 seed	No new orders once settled profit reaches −$50 (the seed is gone), as League already does (§7).

1 Summary

The goal is a model that prices a Valorant match *before each map starts* well enough to beat Kalshi's prices after fees. Everything else (pages, live in-game odds) waits until that is proven on paper trades.

- **Core model: player-based round Elo.** A Valorant map yields about 22 rounds, not one bit, so rate teams on *round share* and turn round odds into map odds with an exact first-to-13 calculation (Fig. 3). Ratings belong to players, as in Prometheus, because Valorant rosters churn hard every off-season and players move up from Challengers.
- **Map layer.** Team-by-map deviations, shrunk toward zero, decaying with time and reset on roster change. Side (attack/defense) barely moves pre-map odds (§5.2), so it waits for the in-game model.
- **Veto model.** Kalshi's map markets open about two days ahead and are named by slot ("map 1", "map 2"), not by map. Before the veto, a slot's price is a mix over which map it will be. A veto model gives that mix and the series price. After the veto, the slot becomes one known map.
- **Market blend and Kelly sizing** carry over from Prometheus unchanged in form: blend our logit with the market midpoint, trade only after-fee edge, cap exposure per fixture across correlated series and map markets.
- **Proof before money.** Same rules as Prometheus: paired-bootstrap benchmark gains, a locked holdout (games from 2026-07-01), then six-plus weeks of paper trading at VCT 2027 Kickoff before live, small-cap trading.

Key numbers

	Series (KXVALORANTGAME)	Map (KXVALORANTMAP)
Events listed	1,673 (from 2025-09)	3,316 (from 2026-02)
Tier 1 + EWC	753 · median 115k contracts	1401 · median 25k
Challengers / VCL	669 · median 22k	1397 · median 3.4k
Game Changers	187 · median 30k	403 · median 4.7k
Other third-party	64 · median 8.5k	115 · median 2.1k
Map slots listed		map 1: 1,579 · map 2: 1,579 · map 3: 118 · map 4: 40

Volumes are contracts traded per event (both sides summed), from Kalshi's public and historical market endpoints, pulled 28 Oct 2026. Tiers are classified from each market's rules text.

2 What carries over from Prometheus

Prometheus already solved the hard, unglamorous parts: honest benchmarks, Kalshi matching, fee-aware sizing and a release-controlled executor. Recommend a **new repo with the same conventions** (uv, SQLite rebuilt from scratch, raw SQL, steering docs, work log, ruff, pytest), copying the Kalshi and execution modules first and extracting a shared package only once both projects use them.

Piece	Prometheus source	Valorant use
Player-based Elo with RD and league offsets	prometheus/elo.py (EloState, Glicko-style RD, home-league offset)	Same structure; the result becomes round share instead of a game-length-weighted win.
Paired bootstrap by series, holdout discipline	prometheus/evaluation.py, tech.md Metric change rules	Unchanged. Resample whole series; sections become tier 1 / international / Challengers / GC.
Kalshi catalog, candles, fees	prometheus/markets.py, prometheus/live/kalshi.py	New series tickers; keep the day-slip matching fix.
Market blend	prometheus/market_blend.py, scripts/fit_market_blend.py	Refit per market type (series vs map slot).
Kelly sizing and executor	prometheus/execution/sizing.py, deploy/execution	Same kelly_order and caps; new fixture-level cap across series + maps.
Alert contract, paper ledger	prometheus/alerts.py, alert_contract.py, trading.py	Same freeze-then-settle pattern for paper trades.
Steering docs, work log, CI budget	AGENTS.md, docs/steering/*	Copy the conventions.

What is genuinely new

LoL (Prometheus)	Valorant
Oracle's Elixir CSVs: one clean download	No official API; scrape VLR.gg pages and keep a raw cache
A game is one result	A map is ~22 rounds: far more signal per map
Draft happens in-game	Map veto happens before play and decides which maps are played
One map (Summoner's Rift)	Pool of 7 maps that rotates during the year; team strength varies by map
Kalshi lists series and games	Kalshi lists series and map *slots* (map 1, 2...), open before the veto

3 The market

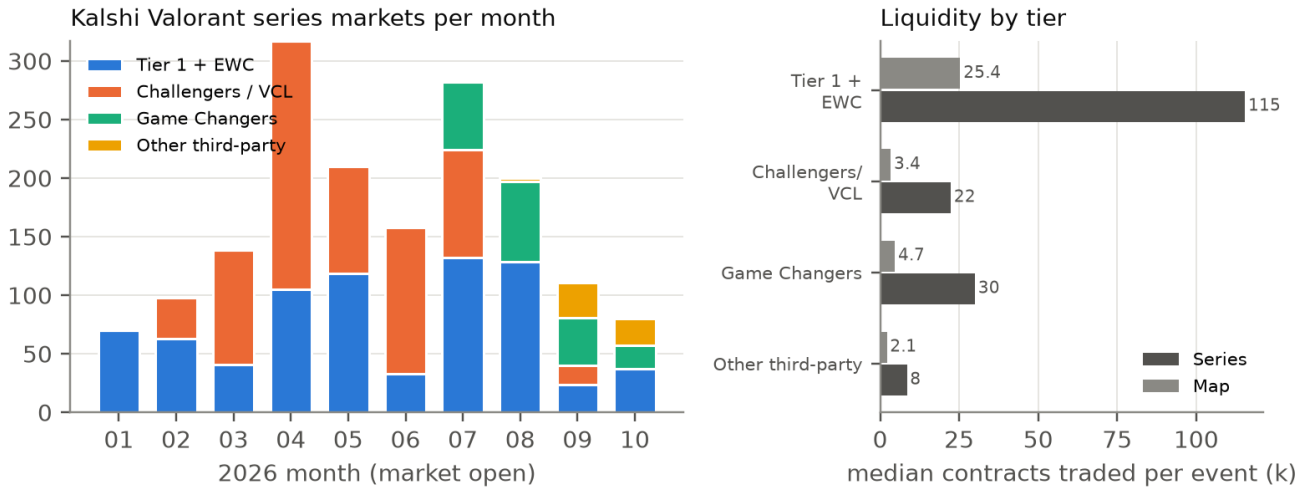


Fig. 2. Left: Kalshi Valorant series events by month of market open (8 more in Sep 2025 not shown). Right: median contracts traded per event. Tier 1 carries the money; Challengers/VCL has volume of listings but about a fifth of the liquidity.

Findings that change the design

- **Map markets are slots, not maps.** "KRÜ Blaze wins map 2" opened on 28 Oct for a 30 Oct match, two days before anyone knows which map is map 2. The veto turns a mixed bet into a known one, and that information shock is the clearest pre-map edge to test.
- **Maps 1 and 2 are nearly all of it.** Map 3 is listed for only 118 events and map 4 for 40, against 1,579 for map 1. Design around maps 1–2 and the series.
- **Unplayed maps settle at "fair market price"**: a map, match or forfeit cancelled before play resolves to the fair price, not to No. A map-3 position in a 2–0 isn't a loss; model it as a return to the market price.
- **Settlement sources** are valorantesports.com, eGamersWorld and bo3.gg, so cross-check results against bo3.gg.
- **Stable team IDs.** Each market carries `custom_strike.esports_competitor`, a UUID per team. Map it once in an alias table instead of fuzzy-matching names like "MIBRF".
- **Fees.** Prometheus uses a taker fee of $0.07 \times P \times (1 - P)$ per contract and fetches each series' rate. Confirm the Valorant series' rates the same way.

Liquidity sets the bet size. A median tier-1 map event trades about 25k contracts; a Challengers map about 3.4k. Cap each order at a small share of displayed depth, and expect tier 1 to carry most of the profit even though Challengers has more listings.

4 Data

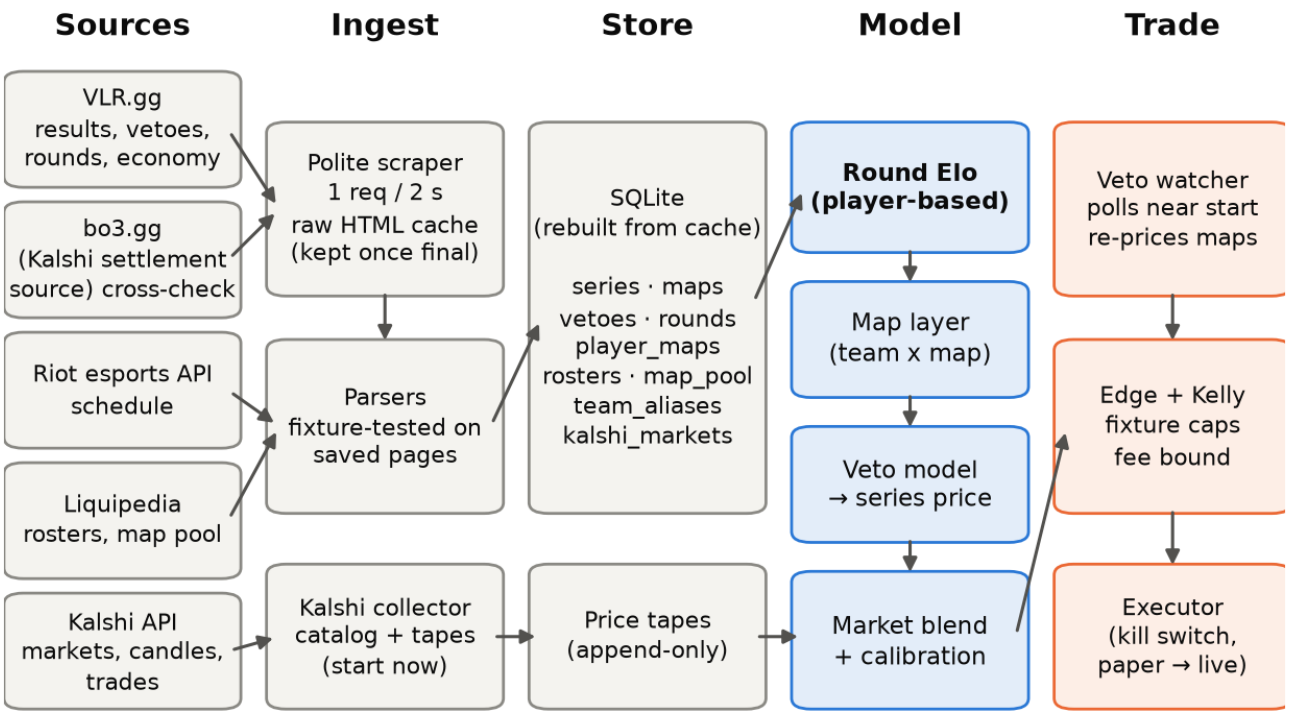


Fig. 1. Pipeline. The DB is always rebuilt from the raw cache, as Prometheus rebuilds from CSVs. The Kalshi collector starts first because order books can't be backfilled.

4.1 Sources

Source	What it gives	Notes
VLR.gg match pages	Veto line, map scores, attack/defense halves, round-by-round win type, Economy tab, per-player rating/ACS/KAST/ADR/FK/FD, agents, patch, date	Primary. Unofficial; no API. Scrape slowly, cache raw HTML forever (completed matches don't change), parse from cache. Verify economy fields in Phase 1.
bo3.gg	Results, maps, schedule	A Kalshi settlement source. Use it to cross-check scores and as a fallback if VLR breaks.
Riot esports schedule	Official schedule for VCT and GC	Likely the same persisted API as lolesports (getSchedule). Verify the key and endpoints.
Liquipedia (MediaWiki API)	Rosters with dates, map-pool rotation history, team renames	Strict rate limits and attribution (CC BY-SA). Fetch rarely, cache.
Kalshi API	Market catalog, hourly candles, trade tapes, settlement	Public market data needs no key. Start collecting now; tapes are append-only.
Oracle's Elixir Valorant (not used)	Per-tournament player and team stats, RAR (GRID data)	No bulk or CSV download (the site's routes are stats, teams, players and RAR); no veto or round rows; data offered for non-commercial use. A sanity check for tier-1 player stats at most.

4.2 Tables

One numbered SQL script per step, as in Prometheus. Main tables:

Table	Grain and key columns
events	event_id, name, tier (tier1 / international / challengers / ascension / gc / third_party), region, LAN flag
series	series_id, event_id, start_utc, best_of, team_a, team_b, winner, patch, VLR / bo3 / Kalshi ids
vetoes	series_id, step, team, action (ban / pick / decider), map: parsed from the VLR veto line
maps	map_id, series_id, order, map, picked_by, rounds_a, rounds_b, first-half side, half scores, overtime rounds
rounds	map_id, round_no, winner, winner_side, win_type (elim / spike / defuse / time), buy types and bank if available
player_maps	map_id, player_id, team, agent, rating, ACS, KAST, ADR, FK, FD (the five starters = roster for the rating)
map_pool	map, active_from, active_to (veto model restricts to the live pool)
team_aliases	team_id, VLR name, bo3 name, Kalshi code, Kalshi competitor UUID, valid_from/to
kalshi_markets / prices	ticker, event, kind (series / map n), team_id, open/close, result; candles and trades

4.3 Backfill and checks

- **Scope:** all VCT-era pro matches from 2023 on (the franchised era), plus 2021–22 to seed player ratings. Tens of thousands of maps. At 1 request per 2 s and about 2 pages per match, a full backfill is roughly a day or two of polite crawling, run once.
- **Parser tests** on saved pages for every format (Bo1, Bo3, Bo5, overtime, forfeits, stand-ins, missing economy).
- **Integrity checks:** map winners add up to the series score; rounds add up to map scores; every Kalshi-listed match maps to one series; results agree with bo3.gg.
- **Terms:** respect robots.txt and a low request rate, set a contact User-Agent, and never re-fetch a cached final page. If VLR objects or blocks, the adapter layer swaps in bo3.gg or a paid feed (GRID, PandaScore).

5 The model

Five layers, each one a candidate that must earn its place on the benchmark (§6). A and B are the Elo model you asked for; C turns map odds into series and map-slot prices; D and E make the price trade-ready.

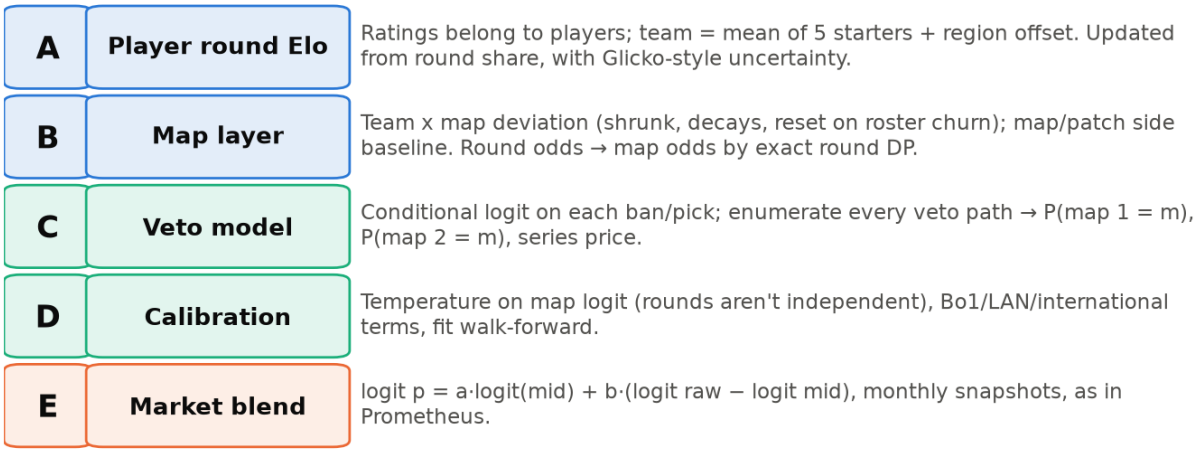


Fig. 4. The model stack. Blue: ratings. Green: pricing and calibration. Orange: the market layer.

5.1 Layer A: player-based round Elo

Each player has a rating R and an uncertainty RD. A team's strength on a map is the mean of its five starters' ratings plus its region's offset (as in Prometheus). For team A against team B on map m:

$$\Delta = (\text{mean } R_A + \text{off}_A) - (\text{mean } R_B + \text{off}_B) + \delta_{A,m} - \delta_{B,m}$$
$$e = \sigma(\Delta / s) \qquad \text{expected share of rounds A wins}$$

after the map, for each starter i on A (mirror for B):

$$R_i \leftarrow R_i + K \cdot (RD_i^2 / RD_{floor}^2) \cdot (w_A/n - e)$$

w_A/n is the share of rounds A won, so a 13–3 moves ratings far more than a 14–12. That is the Valorant counterpart of Prometheus's game-length-weighted "actual" score. RD grows with time away and shrinks with play, so a new or returning player moves fast. Region offsets move only on cross-region games: Masters, Champions, EWC, Ascension, GC Championship and third-party internationals. Treat these like Prometheus's `INTERNATIONAL_LEAGUES`; a missing event silently splits the pools.

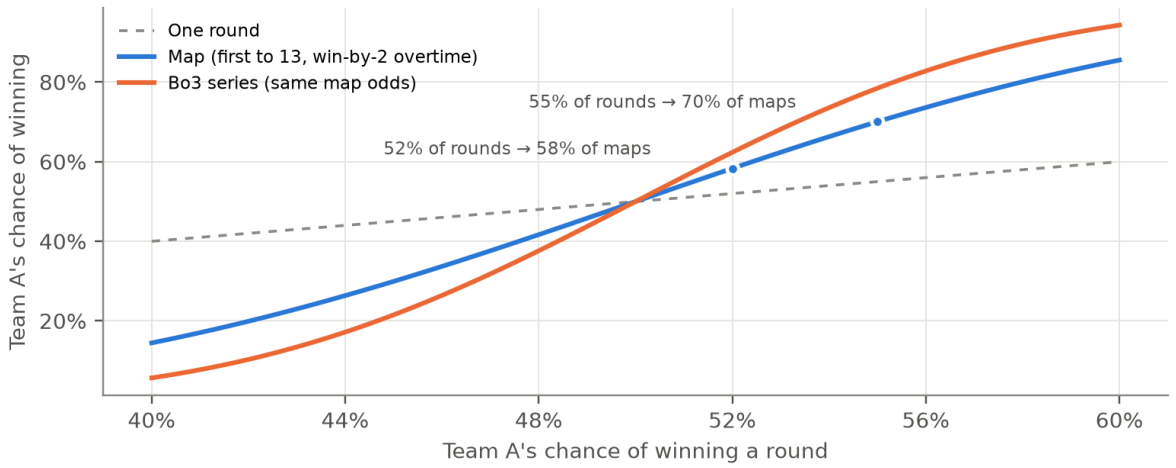


Fig. 3. Small round edges become big map edges. With equal odds on both sides, a 52% round edge is a 58% map favorite and 55% is 70%. This assumes rounds are independent, which they aren't (economy carries over), so Layer D shrinks the result.

Why players, not teams. Valorant teams rebuild every off-season, use stand-ins, and promote players from Challengers. A player-based rating keeps what each player has shown. A team-based rating would reset or carry stale strength. The benchmark still tests a team-based version as a baseline.

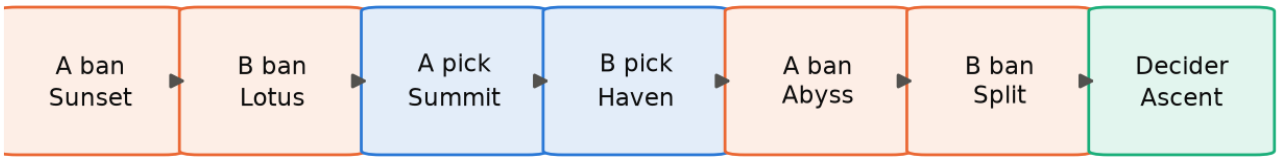
Separate pools that link. Game Changers players rarely face tier-1 players, so GC forms a mostly separate pool linked through the GC Championship and mixed events. Give it its own offset and check its calibration separately.

5.2 Layer B: map strength (and why side can wait)

$\delta_{T,m}$ is a team's extra round-win log-odds on map m: the average residual of its recent maps there, shrunk by a prior weight k (δ = Σ residuals / (n + k)), decayed with a half-life, and scaled by the share of current starters who played those maps. Maps new to the pool start at $\delta = 0$ with wider uncertainty. Fit k, the half-life and the roster scaling on the tuning period only.

> **Finding: starting side barely matters for pre-map odds.** If a team wins 47% of attack rounds and 57% of defense rounds, its map odds are 58.4% whichever side it starts on, nearly identical to a flat 52% (58.3%). Each team plays 12 rounds per side, so side bias averages out. Model per-map side baselines only for the in-game model later; don't spend pre-map effort on side choice.

5.3 Layer C: the veto model



KRÜ Blaze (A) vs Team Liquid Brazil (B), Game Changers Championship, 28 Oct 2026, from VLR.gg.
Picks become maps 1 and 2; the decider is map 3.

Fig. 6. A real Bo3 veto. Bo3 is ban, ban, pick, pick, ban, ban, decider; Bo5 is ban, ban, then four picks and a decider. With 7 maps, either format has 7! = 5,040 possible orders, few enough to list every one exactly.

Model each step as a conditional logit over the maps still available: the chance team T bans or picks map m rises with T's map-strength edge there ($\delta_{T,m} - \delta_{opp,m}$, reversed for bans), T's shrunk historical rate of picking or banning m, and how recently T has played it. Fit on every parsed veto in the tuning period.

Then for each match, enumerate all veto orders with their probabilities:

- **Map-slot price before the veto:** P(A wins map 1) = Σ_m P(map 1 = m) · p_m, where p_m is A's map-win probability on m.
- **Series price:** Σ over veto orders of P(order) · Bo3(p_{map1}, p_{map2}, $p_{decider}$), with Bo3(a, b, c) = ab + a(1−b)c + (1−a)bc.
- **After the veto:** the slots are known, so the prices become plain p_m and the series price uses the three known maps.
- Even a crude veto model (map frequencies only) beats a map-blind price; benchmark the veto log loss on its own as well.

5.4 Layer D: calibration

Rounds aren't independent: economy, pistols and momentum make round results cluster, so the round-DP map odds are too confident. Fit a temperature τ on the map logit ($p = \sigma(\tau \cdot \text{logit } p_{DP})$), walk-forward, plus small terms for Bo1 vs Bo3, LAN vs online, and cross-region games (as in Prometheus's regional calibration). An economy-adjusted round share (discounting anti-eco wins) is a research candidate once the economy data is verified.

5.5 Layer E: market blend

logit p = a · logit(mid) + b · (logit p_{model} − logit mid), fit monthly on earlier settled markets, separately for series and map slots, as in Prometheus's `market_blend.py`. The fitted b tells you how much the market under-uses the model: b ≈ 0 means no edge.

5.6 The candidate ladder

#	Candidate	Adds	Must beat
M0	Coin flip / tier-1 baseline	Floor	-
M1	Team Elo on map wins	Simplest real model	M0
M2	Team Elo on round share	Margin information	M1
M3	Player round Elo + RD + region offsets	Roster churn, cross-tier players	M2
M4	+ map layer (δ)	Map strength	M3 on map log loss
M5	+ veto model	Pre-veto map-slot and series prices	M4 with a frequency-only veto
M6	+ calibration terms	Bo1/LAN/region, temperature	M5
M7	+ market blend	Market information	Kalshi midpoint alone

6 Proving it

- **Same rule as Prometheus:** change the model only for a significant gain. The paired-bootstrap 95% interval of the log-loss difference, resampling whole series, must exclude 0, with no guarded section (tier 1, international, Challengers, GC) significantly worse. An added parameter also needs ≥ 0.0005 log loss.
- **Locked holdout:** tune and select on maps before 2026-07-01. 2026-07-01 onward (Champions, EWC, GC Championship and 673 Kalshi-listed series so far) is scored once per model family. The real test is

paper trading from Kickoff 2027.

- **Score three things separately:** map winner (post-veto), map-slot (pre-veto) and series (pre-veto). Report log loss, Brier and calibration by tier.
- **Against the market:** at fixed times (T−12 h, T−1 h, after the veto) compare our price with Kalshi's midpoint and ask from hourly candles and trade tapes. Track simulated after-fee ROI and **closing-line value** (does our side's price move toward us before the start?). CLV is the earliest honest signal of edge.
- **Momentum check:** after map 1, do Kalshi's map-2 prices move more than the model says they should? That would make the between-map window a second edge.

Gates before real money

Gate	Pass condition
Model	M3+ beats M1 significantly on map log loss in the tuning period, then holds up on the holdout
Market (holdout)	Blend coefficient b significantly > 0, and simulated after-fee ROI positive at the trade times used, on executable asks
Paper (Kickoff 2027)	≥ 6 weeks and ≥ 150 paper trades; positive CLV with a 95% interval above 0; ROI interval reported
Ops	Veto watcher latency measured; kill switch tested; every Kalshi Valorant market maps to one series

7 Trading

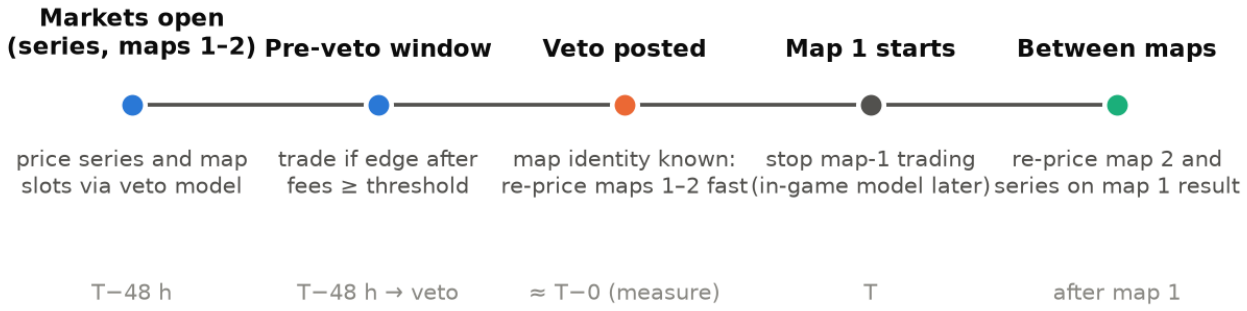


Fig. 5. The pre-map trading windows for one match. The veto window is the one to measure first: how long after the veto appears on VLR or the broadcast do Kalshi's slot prices move?

- **Edge** = p_{blend} − ask − fee, with fee = 0.07 · ask · (1 − ask) per contract (confirm per series). Example: p = 0.60 at a 0.55 ask pays a 0.0173 fee for a 0.0327 net edge.
- **Sizing:** Prometheus's `kelly_order`. Start at quarter Kelly until 300 settled trades, then revisit with a sizing simulation as Prometheus did.
- **Correlation:** series, map 1 and map 2 on one match are the same bet several times over. Cap total exposure per fixture across all its markets, and per day.
- **Depth:** cap each order at a small share of displayed depth. Skip markets whose spread is wider than the edge.
- **Veto watcher:** from T−60 min, poll the VLR and bo3.gg match pages every 20–30 s. On a parsed veto, re-price the slots and series and place orders within seconds. Log the veto's first-seen time

against Kalshi's trade tape to measure the window.

- **Between maps:** after map 1 ends, re-price map 2 (map known) and the series. Same guards; stop at map start.
- **Controls:** paper mode first, an off switch independent of the code, and risk limits in config, all as in Prometheus's executor.
- **Limits:** the same as League: $250 per order and per fixture, $1k exposure, $2k per day, and the same Kelly settings, from a **$50 seed**.
- **Stop at −$50:** capital = seed + settled profit − open stakes; no new orders when it is $0 or less, so trading stops once settled profit reaches −$50. Prometheus's executor already enforces this (prometheus/execution/core.py); reuse it unchanged.

8 Repo layout (proposed)

```
valorant-model/                  # working name
  AGENTS.md  docs/steering/        # same steering conventions as Prometheus
  promval/ingest/                  # vlr.py, bo3.py, riot_schedule.py, liquipedia.py, cache.py
  promval/parse/                   # series, veto, maps, rounds, economy parsers
  promval/ratings/elo.py           # player round Elo, RD, region offsets (from prometheus/elo.py)
  promval/ratings/maps.py          # δ team x map layer
  promval/pricing.py               # round DP, Bo1/3/5, veto enumeration
  promval/veto.py                  # conditional-logit veto model
  promval/markets/                 # Kalshi catalog, tapes, team-id matching, blend
  promval/trading/                 # edge, kelly sizing, paper ledger, executor client
  scripts/0NN_*.sql|py             # DB build in numeric order
  scripts/evaluate_*.py            # benchmarks: model, veto, market
  tests/fixtures/vlr/*.html        # saved pages for parser tests
```

9 Roadmap

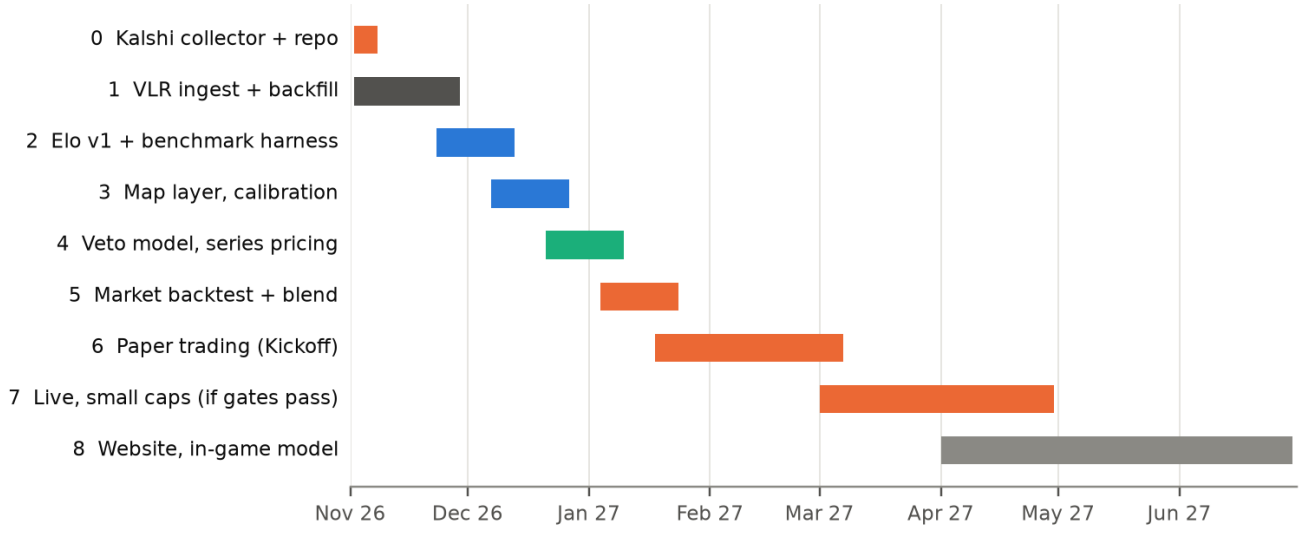


Fig. 7. Proposed schedule. The VCT 2027 season typically starts with Kickoff in mid-January, so the model should be ready to paper trade by then.

Phase	Deliverable	Exit criterion
0	Repo; Kalshi collector for all Valorant markets (catalog, candles, tapes) running daily	Every active Valorant market's tape saved; alias table seeded from competitor UUIDs
1	VLR scraper, cache, parsers, DB build; bo3 cross-check	Backfill complete; integrity checks pass; 95%+ of Kalshi matches joined to a series

Phase	Deliverable	Exit criterion
2	M0–M3 and the benchmark harness (bootstrap by series, sections, holdout lock)	Report with M3 vs M1 numbers
3	Map layer and calibration (M4, M6)	Significant map log-loss gain or a recorded rejection
4	Veto model and series/slot pricing (M5)	Veto log loss vs the frequency baseline; pre-veto series and slot scores
5	Market backtest and blend fit	Holdout ROI/CLV report; go or no-go for paper trading
6	Paper trading with the veto watcher	Paper gate (§6)
7	Live trading, small caps	Weekly review; kill switch drills
8	Website and in-game model (later)	-

10 Risks

Risk	Mitigation
VLR blocks scraping or changes its HTML	Raw cache (never re-fetch), parser tests on saved pages, bo3.gg adapter, paid-feed fallback budgeted later
Thin markets: edge exists but can't be filled	Depth caps, focus on tier 1, measure fills in paper mode against tapes
Veto window too short or nonexistent	Measure it first (Phase 0 tapes + watcher logs); the pre-veto model stands on its own anyway
Off-season roster churn	Player-based ratings with RD; roster-scaled map layer
Map-pool rotation and patches	map_pool table; new maps start neutral with wider uncertainty; patch as a calibration term if it earns it
Pools that don't link (GC, small regions)	Own offsets; calibration by section; all cross-region events registered
Overfitting to a short Kalshi history	Holdout locked; blend fit only on earlier months; paper trading before money
Mis-matched markets (renames, day-off listings)	Competitor UUIDs; the Prometheus day-slip fix; ambiguous matches give nothing

11 Decisions

Decision	Status
Backfill depth	Decided: score 2023 on; 2021–22 seeds player ratings
Third-party events	Decided: in the ratings and traded
Bankroll and caps	Decided: League's limits, $50 seed, stop at −$50 settled profit (League already does this)
Data source	Decided: VLR.gg primary, bo3.gg cross-check. Oracle's Elixir Valorant has no bulk download
Veto watcher host	Recommended: a 1-minute scheduled Lambda that exits at once outside match windows, not the EC2 worker (§7)
Repo	Open: a new repo (recommended) or a package inside Prometheus, and its name

Why not the EC2 worker for the veto watcher. The worker isn't always on: it powers off after 90 idle minutes, and Prometheus's hourly prices job wakes it about 15–75 minutes before a LoL match. Using it for Valorant would need a second wake job in the new repo, permission for that repo to start the instance, and an hourly wake that GitHub's cron can delay. A scheduled Lambda is always ready, costs almost nothing when idle, and already matches how the pre-match executor runs (a scheduled Lambda).

Appendix: the map-odds calculation

Let a and d be A's round-win chances on attack and defense, and A start on side s. A map is first to 13 rounds; sides swap after round 12; at 12–12, overtime is played in pairs (one round per side) until one team leads by two. With f(i, j) the chance A wins from score i–j:

```
f(13, j) = 1,  f(i, 13) = 0
f(12, 12) = a·d / (a·d + (1−a)(1−d))      (overtime)
f(i, j)  = p · f(i+1, j) + (1 − p) · f(i, j+1),  p = a or d by the side A plays in round i+j+1
```

Assumes rounds are independent given sides; Layer D corrects for the clustering. Sources: Kalshi trade API (series KXVALORANTGAME, KXVALORANTMAP; public and historical market endpoints), VLR.gg match pages, Prometheus code and steering docs (main at b896d93).